

OPPORTUNITIES AND CHALLENGES IN THE DEVELOPMENT OF THE DIGITAL ECONOMY IN VIETNAM

Huy Thanh Phan¹, Hong Van Luong², Dieu Anh Bui³

¹Ministry of Finance

²CMC University

³National Economic University

Abstract: *In the context of global acceleration toward digital transformation, Vietnam has designated the digital economy as one of the three pillars of its National Digital Transformation Strategy to 2025, with a vision to 2030. This paper examines the opportunities and challenges shaping Vietnam's digital economy, drawing on GSO data (2020–2024), international benchmarks, and comparative evidence from China, India, Indonesia, and Malaysia. Opportunities stem from political commitment, rapid platform growth, a young and tech-savvy population, and the emergence of local technology firms. At the same time, persistent challenges remain, including pronounced regional disparities in digital capacity, limited investment in R&D, shortages of high-quality digital talent, reliance on foreign platforms, and fragmented policy frameworks. To address these gaps, the paper provides a pillar- and province-level diagnosis and proposes targeted measures to strengthen domestic platform development, enhance digital governance, and build human capital, thereby advancing an inclusive and sustainable digital economy toward 2030.*

Keywords: *challenges; digital economy; opportunities; policy solutions; Vietnam.*

1. Introduction

The digital economy has emerged as a central engine of growth and innovation worldwide, reshaping production structures, labor markets, and public policy. Economies that successfully navigate digital transformation not only achieve productivity gains but also strengthen international competitiveness by developing indigenous technology ecosystems and large-scale digital platforms. Against this backdrop, Vietnam has placed digital development at the heart of its national strategy, enacting a range of policies and initiatives to expand connectivity, digital services, and enterprise digitalization (GSO; e-Conomy SEA).

Nevertheless, Vietnam's digital transition faces systemic constraints. Despite improvements in internet penetration and a rising contribution of digital activities to GDP, composite indicators reveal persistent weaknesses in innovation capacity: public and private R&D investment remains low, the supply of high-quality digital talent is insufficient to support platform scaling, and international rankings suggest limited readiness in certain network and institutional pillars. These shortcomings—together with pronounced spatial disparities across provinces—raise the risk of continued reliance on foreign platforms and cloud services, which have implications for data sovereignty and the development of domestic digital value chains (World Bank; WEF NRI; e-Conomy SEA).

Motivated by these observations, this paper addresses the following core research question: What are the principal strengths, weaknesses, and institutional or technical barriers shaping Vietnam's digital ecosystem, and what concrete policy measures could enhance the domestic

capacity to develop “Made-in-Vietnam” digital platforms? The study goes beyond descriptive aggregation for 2020–2024 by decomposing composite indices (such as the pillars of the Network Readiness Index— infrastructure, human capital, institutions, market, and adoption) to pinpoint core bottlenecks.

The contribution of this paper is threefold. First, it systematizes and analyzes quantitative and qualitative evidence for the period 2020–2024 to provide an updated empirical picture of Vietnam’s digital ecosystem. Second, it performs component-level analysis to link inputs (R&D intensity, skills, infrastructure) with outcomes (platform dependence, enterprise digital adoption), thereby strengthening inferential understanding of the observed constraints. Third, it formulates actionable policy recommendations — including financing mechanisms for regional data infrastructure, incentives to increase R&D and platform scaling, and targeted measures to reduce spatial disparities — aimed at enhancing indigenous digital capabilities over the 2025–2030 horizon. The remainder of the paper describes data and methods, presents the current state of the digital economy, analyzes key findings, and concludes with policy implications. (GSO; World Bank; WEF NRI; e-Conomy SEA).

2. Theoretical overview

The concept of “digital economy” began to take shape in the late 20th century, when information and communication technology (ICT) created profound impacts on socio-economic activities. However, it was not until the first two decades of the 21st century that the digital economy truly became a dominant global trend, driven by breakthrough advances in technologies such as big data, artificial intelligence (AI), the Internet of Things (IoT), cloud computing, and the proliferation of digital platforms. According to the Organisation for Economic Co-operation and Development (OECD), the digital economy is defined as “the totality of economic activities that use digital data and digital technologies as primary inputs, related to the digitization of products, services, processes, and business models.” The World Bank (2021) further emphasizes the central role of data, considering it a new form of productive asset in the modern digital economy era.

In essence, the digital economy is not an isolated sector but rather a pervasive component capable of impacting all aspects of the economy—from production processes and distribution systems to consumer behavior and government administration. The distinctive characteristics of the digital economy include: (i) digitization of processes and services; (ii) the use of data as an essential input factor; (iii) the new intermediary role of digital platforms; and (iv) rapid scalability through intangibility and global connectivity capabilities. The growth model in the digital economy increasingly depends on the capacity to collect, process, and analyze big data, as well as the development of the platform economy, digital services, and digital innovation activities. The digital economy brings numerous positive impacts, including enhanced labor productivity through automation and operational optimization; expanded market access through e-commerce and cross-border services; and the promotion of innovative business models and creation of new employment opportunities. However, according to UNCTAD (2019), it also poses significant challenges regarding digital inequality, personal data protection, cybersecurity, and the regulatory issues of digital monopolies by large platforms.

In the context of strong technological globalization, many developing countries have proactively positioned the digital economy as a strategic driver for national development. India serves as a prominent example through its India Stack initiative, a comprehensive digital ecosystem that integrates a universal digital identity (Aadhaar), digital payments, digital finance, and open data

platforms to support public services. With over 1 billion citizens possessing digital identities, India has successfully accelerated the rapid development of fintech companies, digital public services, and e-commerce, while significantly narrowing the gap in banking service access for low-income populations (World Bank, 2023).

China has implemented strategic investments in advanced digital infrastructure, particularly 5G networks, large-scale national data centers, and artificial intelligence research and development. Since 2015, the Chinese government has deployed the "Internet Plus" strategy, aimed at deeply integrating digital technology into key sectors such as agriculture, healthcare, manufacturing, and public administration. Leading technology conglomerates such as Alibaba, Tencent, and Huawei play a leading role in the domestic digital ecosystem while expanding their influence to international markets. Concurrently, China has also strengthened control and regulation of cross-border data to protect data sovereignty and national interests.

In the Southeast Asian region, Indonesia has emerged as the fastest-growing digital economy market. With an abundant young population, a growing middle class, and high internet penetration rates, the country has witnessed an explosion in e-commerce, sharing services, and digital finance. The Indonesian government announced a national digital transformation roadmap in 2020 and prioritized investment in digital infrastructure in remote and rural areas. Similarly, Malaysia has implemented the "MyDigital" policy with the objective of developing national data platforms, accelerating public sector digitization, and encouraging small and medium enterprises (SMEs) to undergo digital transformation through financial support packages and specialized skills training.

When compared with the aforementioned countries, Vietnam has been moving in the right direction by promulgating the National Digital Transformation Strategy to 2025, with a vision to 2030, which identifies the digital economy as one of three main pillars. However, Vietnam still faces considerable gaps in many aspects. According to the Network Readiness Index (NRI) 2023, Vietnam ranks 57th out of 134 countries, significantly lower than Malaysia (38th) and Thailand (43rd).

Notably, according to the United Nations E-Government Survey 2024, Vietnam has achieved remarkable progress in digital government development. The country has risen to 71st position out of 193 countries, advancing 15 places compared to 2022 (and 2020, both ranked 86th). This marks the first time Vietnam has been classified in the "Very High EGDI group" for e-government development, with an EGDI score of 0.7709 in 2024. Within the Southeast Asian region, Vietnam currently ranks 5th out of 11 countries, advancing one position compared to 2022. However, countries such as Singapore (3rd globally), Thailand, Malaysia, and Indonesia still maintain higher rankings.

This indicates that Vietnam needs to build a more synchronized and comprehensive ecosystem encompassing institutions, infrastructure, human resources, and innovation activities to bridge the current gap and effectively capitalize on the opportunities that the digital economy presents.

3. Data collection & Methodology

3.1. Research design

This study is a desk-based policy analysis that synthesizes secondary data and comparative indicators to identify strengths, weaknesses, and institutional/technical barriers of Vietnam's digital ecosystem over 2020–2024. The aim is not to establish rigorous causal inference from microdata, but to systematize quantitative and qualitative evidence, perform component analysis of composite

indices, and develop inferential linkages between inputs (R&D, skills, infrastructure) and outcomes (platform dependence, enterprise digital adoption).

All data for this study are drawn from high-quality secondary sources and industry reports to ensure transparency and verifiability. Macro indicators and provincial breakdowns (digital economy share, GRDP by province, and select enterprise adoption measures) are sourced from the General Statistics Office (GSO). Cross-country contextual data and thematic reports are taken from the World Bank and market estimates from e-Conomy SEA (Google–Temasek–Bain). Component indicators and readiness scores are referenced from the WEF’s Network Readiness Index (NRI) and UNDESA’s EGDI for pillar decomposition where available. Additional evidence on SME digital transformation, labor supply and sectoral case studies is drawn from Ministry of Planning and Investment (MPI) surveys, industry associations, and academic/industry reports (e.g., RMIT, Vietnam Internet Association).

3.2. Data analysis

The analysis follows four main steps: (1) descriptive and comparative analysis of 2020–2024 indicators; (2) component decomposition of composite indices such as NRI/EGDI; (3) simple correlation checks to establish inferential linkages between inputs (R&D, skills, infrastructure) and outcomes (NRI, enterprise digital adoption); and (4) spatial analysis to highlight provincial disparities. As the study relies on secondary data, relationship findings are exploratory and inferential; primary data collection is recommended for stronger causal testing.

4. The current state of Vietnam's digital economy

The period from 2020 to 2024 has witnessed significant strides in the formation and development of Vietnam's digital economy. Under the strong direction of the government and the coordinated efforts of ministries, sectors, and localities, the digital economy has gradually affirmed its role as a new growth driver, contributing to increased productivity and the transformation of the national development model.

4.1. Development level and contribution to growth

GDP contribution and stable growth

Table 1. Measurement Results of Digital Economy's Value-Added Contribution to GDP, 2020-2024

Unit: %

	2020	2021	2022	2023	2024
Total	12.66	12.87	12.83	12.87	13.17
1. Agriculture, Forestry, and Fisheries	0.05	0.05	0.05	0.06	0.06
2. Industry and Construction	6.08	6.20	5.90	5.81	5.96
3. Services	6.53	6.62	6.88	7.00	7.15

Source: General Statistics Office (2025)

The proportion of the digital economy's value-added to Vietnam's GDP averaged 12.88% from 2020 to 2024, ranging from 12.66% (2020) to 13.17% (2024). Although there was a slight decrease

in 2022 and 2023 due to a decline in global demand for electronic, computer, and optical products (which account for over 30% of the digital economy's value-added), a strong recovery was observed in 2024, reaching 13.17% thanks to increased orders in related manufacturing sectors.

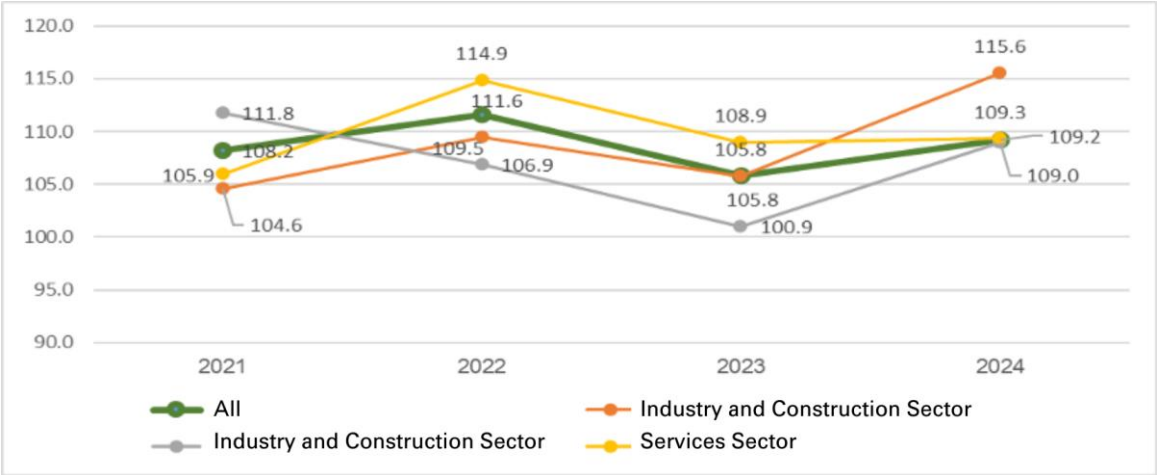
Regarding its structure, the core digital economy contributed approximately 62% (equivalent to 7.99% of GDP), while other digitized sectors accounted for 38% (4.89% of GDP). Notably, the contribution from digitized sectors has shown a gradual increase, from 4.75% in 2020 to 5.01% in 2024, reflecting a deepening trend of digitalization across various economic fields.

Positive growth

In 2024, Vietnam’s digital economy recorded a growth rate of 9.2% compared to 2023, with the core digital economy expanding by 9.9% and digitized sectors growing by 7.5%. Several industries experienced strong growth, including agriculture (over 20%), mining support services (above 30%), coke and refined petroleum production (21%), postal and delivery services (30%), and wholesale and retail trade (over 14%).

On average, during the 2021 – 2024 period, the digital economy grew by 8.7%, led by the services sector (9.7%), followed by agriculture, forestry, and fisheries (8.8%), and industry and construction (7.1%). These figures underscore the pivotal role of the services sector in driving the expansion of the digital economy.

At constant prices, Vietnam’s digital economic activities in 2024 are estimated to have grown by 9.2% compared to the same period in 2023. Within this, the core digital economy expanded by 9.9%, while the digitalization of other sectors rose by 7.5% year-on-year. Several industries recorded notably high growth rates, including: agriculture and related services (over 20%), mining support services (above 30%), coke and refined petroleum production (21%), postal and delivery services (30%), and wholesale and retail trade (over 14%).



Source: General Statistics Office (2025)

Figure 1. Digital economy growth rate by economic sector, 2021–2024 (%)

Scale and digitalization in the services sector

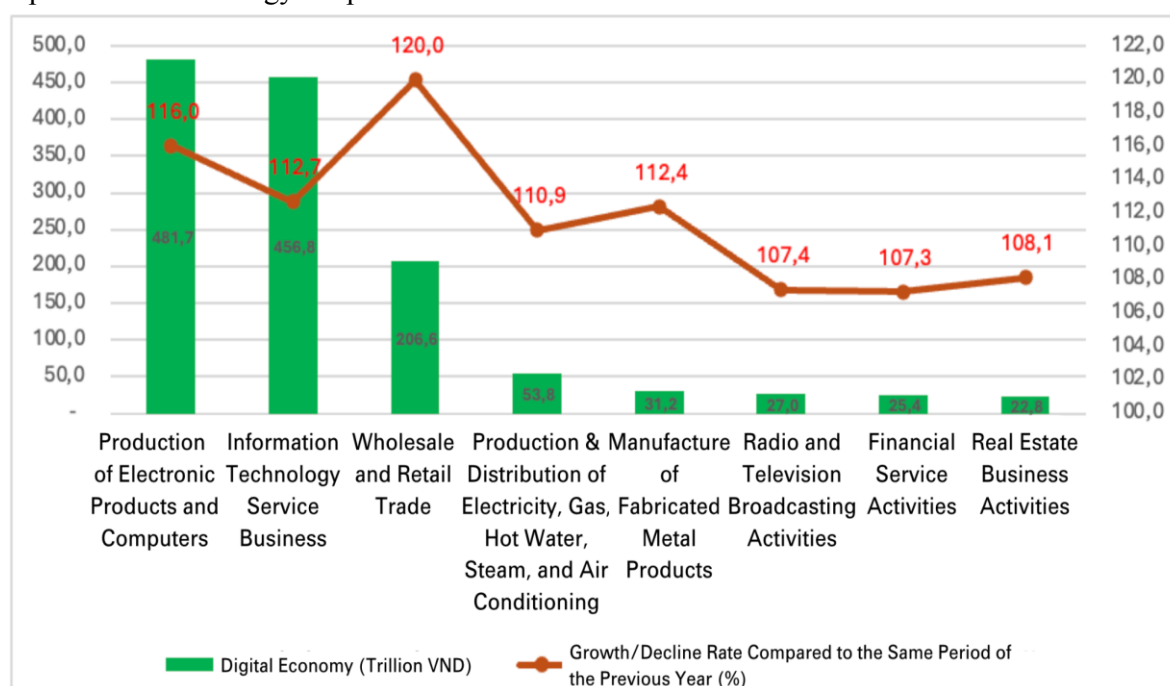
In 2024, the added value of Vietnam’s digital economy was estimated at over VND 1.5 quadrillion (approximately USD 62.7 billion), representing a 14.1% increase compared to 2023 at current prices. The manufacturing of electronic, computer, and optical products contributed more than 32% of this value, growing by 16% year-over-year. Other core digital economy sectors, such

as telecommunications, computer programming, and software wholesaling, accounted for around 30%, with a nearly 13% increase. The total output of the digital economy accounted for 18.7% of the country's total economic output and contributed 49.2% of GDP in 2024. On average, during the 2020–2024 period, these figures were 19.6% and 52.3%, respectively, highlighting the significant role of the digital economy in production and economic value creation.

The value-added generated from IT-enabled service sectors rose from 6.53% of GDP in 2020 to 7.15% in 2024, reflecting the government's and local authorities' efforts to promote digital transformation in governance, operations, and business activities.

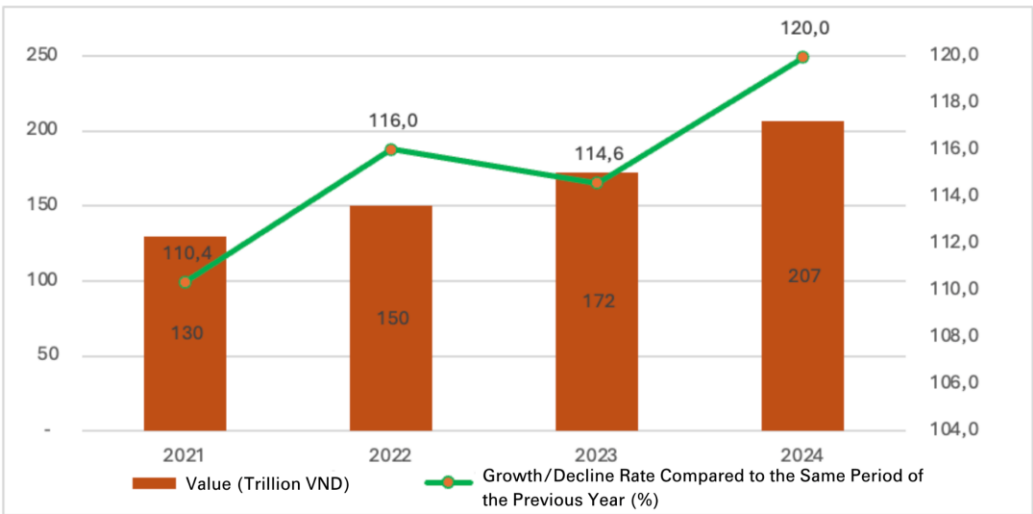
E-commerce (wholesale and retail trade integrated with digital technologies) stood out, contributing nearly 14% of the digital economy's total added value, up 20% compared to 2023. Other sectors also recorded notable growth, including electricity and gas production and distribution (4%, up 11%), finance, broadcasting, and fabricated metal manufacturing (each contributing around 2%).

However, some sectors-such as veterinary services, social assistance, and institutional nursing care-showed extremely low levels of digitalization (approximately 0.002%), revealing significant disparities in technology adoption across industries.



Source: General Statistics Office (2025)

Figure 2. Total value of the digital economy in selected sectors at current prices, 2024



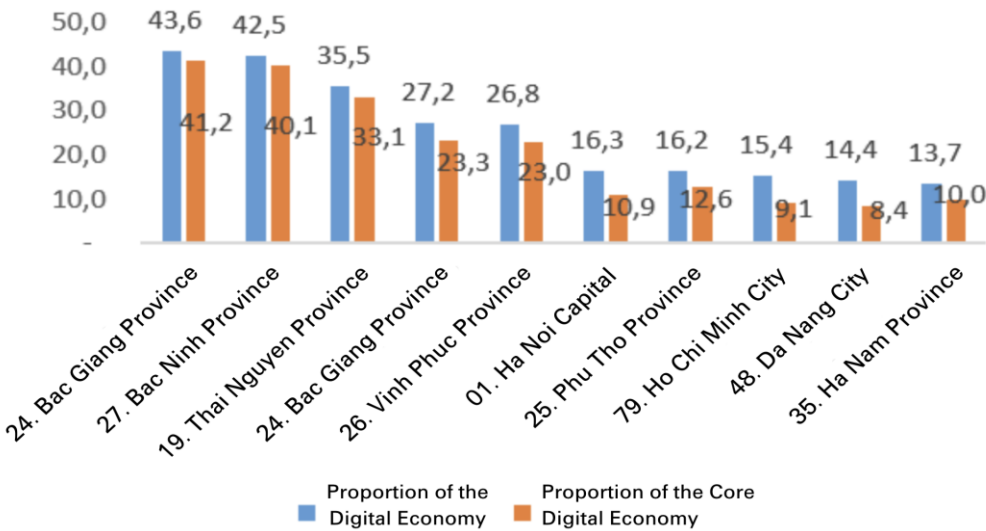
Source: General Statistics Office (2025)

Figure 3. Value of the digital economy in wholesale and retail trade utilizing digital technologies at current prices, 2021–2024

Regional disparities in digital economy contributions

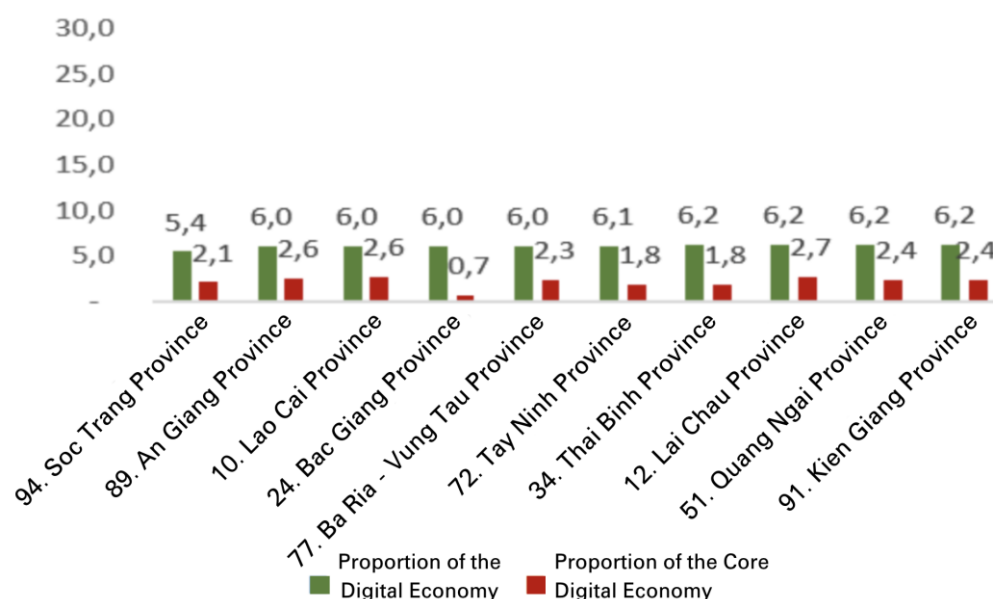
The development of Vietnam’s digital economy has shown clear geographic concentration in a handful of localities. In 2024, five provinces and cities recorded a digital economy share exceeding 20% of their GRDP, including: Bac Giang (43.6%), Bac Ninh (42.5%), Thai Nguyen (35.5%), Hai Phong (27.2%), and Vinh Phuc (26.8%). This concentration reflects the strong growth of the digital economy in regions with advanced technology industries and large-scale electronics manufacturing.

However, the development remains uneven. Only seven provinces and cities reported a digital economy share between 10 – 20%, while the remaining 51 localities fell within the 5 – 10% range. This indicates a significant disparity in digital infrastructure and the degree of technology adoption across regions.



Source: General Statistics Office (2025)

Figure 4. Top 10 provinces and cities by digital economy share of GDP, 2024 (%)



Source: General Statistics Office (2025)

Figure 5. Bottom 10 provinces and cities by digital economy share of GDP, 2024 (%)

Vietnam is widely recognized as one of the most dynamic digital markets in Southeast Asia. According to the 2023 e-Economy SEA report by Google, Temasek, and Bain & Company, the size of Vietnam's internet economy reached an estimated USD 30 billion in 2023, ranking third in the region. Key drivers of this growth include e-commerce, ride-hailing and food delivery services, and digital financial services.

4.2. Digital economy ecosystem and key enabling factors

a) Digital infrastructure

Vietnam has achieved near-universal 4G coverage, reaching 99.8% of the population, and is currently deploying commercial 5G services in major cities such as Hanoi, Ho Chi Minh City, and Da Nang. Despite a high number of internet users—approximately 86 million in 2023, accounting for over 88% of the population—a notable digital divide persists between urban and rural areas.

The development of data centers and cloud computing infrastructure is gaining momentum, with participation from leading domestic technology corporations. However, the cost of cloud services in Vietnam remains approximately 30–40% higher than the ASEAN average, posing challenges for small and medium-sized enterprises (SMEs) in accessing and adopting these technologies.

b) Digital workforce and skills

Vietnam's ICT workforce is estimated at around 1.08 million workers, with over 80% employed in software development and IT service companies. Nevertheless, the World Economic Forum (WEF) reports that the country's general digital skill level remains below the ASEAN average, particularly in areas such as data analytics, cybersecurity, and user experience/interface (UX/UI) design.

While the National Digital Transformation Program aims to train one million high-quality digital professionals by 2030, the implementation of specialized training programs has been slow.

Moreover, there is a lack of effective coordination among government agencies, the private sector, and educational institutions, which continues to hinder progress in building a competitive digital talent pipeline.

c) Digitalization among enterprises

The rate of digital technology adoption among Vietnamese enterprises has been increasing, though primarily at a basic level, with most firms applying digital tools for financial accounting, e-invoicing, and digital signatures. According to the General Statistics Office (2023), only about 21% of businesses use Enterprise Resource Planning (ERP) or Customer Relationship Management (CRM) systems, and fewer than 8% have integrated advanced technologies such as artificial intelligence (AI) or big data into their production and business operations.

The growth of major technology corporations such as Viettel, FPT, VNG, and CMC has served as a key driver for the digital ecosystem. Additionally, the innovation-driven startup ecosystem has expanded rapidly, with over 4,000 startups across the country, some of which have gained international recognition. However, the majority of Vietnamese tech firms still rely heavily on the domestic market and have yet to achieve global competitiveness on par with regional peers.

d) Institutional and policy framework

Vietnam has enacted several strategic policy frameworks to accelerate digital transformation, including the National Digital Transformation Strategy to 2025 with a Vision to 2030, and the National Strategy for the Development of the Digital Economy and Digital Society to 2025, with an Orientation to 2030. Notably, in 2024, the National Assembly passed three landmark legislations: the Data Law, the Digital Technology Industry Law, and the Personal Data Protection Law.

The synchronized introduction of these three laws marks a significant milestone, demonstrating strong political commitment and institutional readiness for digital transformation. These regulations are expected to define, value, and govern both public and private data as strategic assets, thereby enabling the growth of emerging sectors such as FinTech, GovTech, HealthTech, and EdTech, and laying the foundation for a data-driven, domestically empowered digital economy.

4.3. Regional comparison

To assess Vietnam’s position in the digital economy landscape, it is essential to compare its performance with that of comparable or leading countries in the region, including Malaysia, Indonesia, Thailand, China, and India.

Table 2. Comparative indicators of digital economy development: Vietnam and selected countries (2022 – 2024)

Indicator / Country	Vietnam	Malaysia	Indonesia	Thailand	China	India
Digital Economy as % of GDP (2024)	13.17 %	~ 22 %	~ 19 %	~ 18 %	> 40 %	~ 24 %
Digital Economy	49.2 %	~ 55 %	~ 47 %	~ 51 %	~ 65 %	~ 57 %

Indicator / Country	Vietnam	Malaysia	Indonesia	Thailand	China	India
Output (% of GDP, 2024)						
Network Readiness Index (NRI) 2023	57/134	38/134	59/134	43/134	23/134	56/134
Internet Users (% of population, 2023)	88 %	96 %	77 %	85 %	76 %	70 %
R&D Investment (% of GDP, 2021)	0.53 %	1.04 %	0.28 %	1.14 %	2.44 %	0.70 %
E-Government Development Index (EGDI, UN 2022)	Upper-middle	High	Upper-middle	High	Very high	Upper-middle
National Digital ID Scheme	Under implementation	Yes (MyDigital ID)	Under implementation	Yes	Yes (multi-system)	Fully implemented (Aadhaar)

Sources: GSO (2025), World Bank, UNDESA, OECD, WEF, e-Conomy SEA 2023.

Table 2 offers an overview of Vietnam's relative position in the context of regional and global digital economy development, based on comparative indicators from countries such as Malaysia, Indonesia, Thailand, China, and India during the 2022 – 2024 period.

First, regarding the share of the digital economy in GDP in 2024, Vietnam reached 13.17%, indicating a notable contribution, though still significantly lower than major regional economies such as Malaysia (~ 22%), India (~ 24%), and especially China (> 40%). However, when assessed by the share of digital gross output (GO) in GDP, Vietnam reported 49.2%, which is comparable to Indonesia (~ 47%) and Thailand (~ 51%). This reflects the country's strong capacity to integrate digital technologies into production processes, particularly in export-oriented industrial sectors, and represents a key strength in Vietnam's digital development strategy. A closer examination of these indicators indicates that Vietnam's comparative advantage presently resides in production-side digitalization—notably the integration of digital technologies within manufacturing processes and export-oriented value chains—rather than in the endogenous generation of digital platforms. This pattern implies that while firms are effective adopters and integrators of existing technologies, the

country has not yet developed the upstream capabilities (e.g., sovereign cloud infrastructure, platform engineering competencies, and proprietary AI research) necessary to originate scalable, home-grown platforms. Closing this supply-side deficit, therefore requires coordinated policy interventions that shift resources and incentives toward innovation and capability building—such as targeted support for industrial software development, localized cloud and edge investments, and sustained public–private financing of R&D and commercialization pathways—so that applied production strengths can be translated into upstream platform creation.

Second, according to the Network Readiness Index (NRI) 2023, Vietnam ranked 57th out of 134 countries. This demonstrates a degree of progress but still lags behind ASEAN peers such as Malaysia (38th) and Thailand (43rd). Nevertheless, Vietnam outperformed Indonesia (59th) and is on par with India (56th). The NRI is a comprehensive indicator that evaluates the state of digital infrastructure, digital skills, innovation environment, and the societal impact of digital transformation. It thus serves as a critical metric for assessing the potential for sustainable long-term digital economy development. A deeper reading of the NRI ranking requires breaking the index into pillars: Vietnam’s mid-rank typically reflects relatively strong connectivity/adoption but weaker scores on innovation and institutional pillars (policy coherence, R&D ecosystem, startup finance). By contrast, Malaysia and Thailand tend to score better on governance and enterprise readiness, which helps explain their higher overall ranks despite similar internet penetration. Therefore, obtaining and analysing pillar-level NRI scores is essential to pinpoint which institutions would yield the largest gains.

Third, with 88% of the population using the Internet in 2023, Vietnam boasts one of the highest internet penetration rates in the region. This widespread connectivity provides a solid foundation for the expansion of the consumer-facing digital economy, including e-commerce, digital financial services, and cross-border digital platforms—ultimately fostering broad-based public engagement in digital economic activities. However, high penetration does not automatically translate into high-value platform growth: differences in connection quality (fixed vs mobile), urban-rural divides, data affordability, and digital skills determine actual usage intensity. In other words, Vietnam’s demand-side readiness is strong, but to leverage this for platform creation, the country must close gaps in service quality, lower unit costs for platform operators (e.g., cloud hosting), and raise user and enterprise digital literacy so that both consumers and businesses can engage in more complex digital transactions.

Fourth, Vietnam still has significant room for improvement in terms of investment in research and development (R&D) and core technological innovation. With R&D expenditure accounting for just 0.53% of GDP in 2021, the figure falls well below that of China (2.44%) and South Korea (4.81%). This represents a substantial barrier to achieving technological autonomy in key areas such as artificial intelligence (AI), big data, cybersecurity, and the development of locally made digital platforms under the “Made in Vietnam” initiative. At the component level, low R&D intensity translates into weak upstream capabilities - limited domestic development of cloud stacks, platform middleware, and foundational AI models—so Vietnamese firms often remain integrators rather than originators. The policy implication is straightforward: without stepped-up public and private R&D funding, plus stronger university-industry linkages and incentives for commercialization, Vietnam will struggle to scale indigenous platforms and will remain dependent on foreign technology providers for critical infrastructure and core software.

Finally, in terms of digital identity infrastructure, Vietnam is currently in the early stages of deploying its national digital ID system, VNeID, which has yet to achieve universal coverage or deep integration into financial services, public administration, and the broader digital ecosystem. In comparison, India has already rolled out the Aadhaar digital ID to over 1 billion citizens, while Malaysia's "MyDigital ID" initiative is also fully operational. Digital identity is a foundational element and strategic enabler for citizen data management, and its widespread adoption is critical for facilitating broad-based and inclusive digital transformation. From a component perspective, incomplete digital ID coverage undermines secure onboarding, increases KYC friction for fintech and e-commerce, and limits government-to-person (G2P) and public service integration. To enable scalable, trustable platforms - and to lower transaction costs for both consumers and firms - policy should prioritise fast, privacy-aware VNeID integration across banking, payments, health and administrative services, while simultaneously strengthening data protection and interoperability frameworks.

5. Opportunities and challenges

5.1. Opportunities

The development of Vietnam's digital economy is being propelled by a set of strategically favorable factors that provide a robust foundation for transformation and sustainable growth.

First, political and social stability serve as essential prerequisites for the successful implementation of long-term development strategies, including national digital transformation. The Communist Party and the Government of Vietnam have consistently identified the digital economy as one of the three core pillars of the National Digital Transformation Strategy to 2025, with a Vision to 2030, alongside digital government and digital society. Policy coordination and execution-synchronized from central to local levels - have established a clearly defined institutional framework that facilitates an enabling environment and initial policy support for businesses, citizens, and the broader digital ecosystem. This stable and coherent policy orientation reduces policy uncertainty for long-horizon investments (e.g., data centers, R&D facilities) and creates a predictable environment in which domestic firms and foreign partners can commit capital and skills to multi-year digital projects.

Second, Vietnam benefits from a demographic dividend, with a young, dynamic, and tech-savvy labor force that is highly adaptable to technological change. With approximately 86 million internet users in 2023, Vietnam ranks among the countries with the largest digital user bases globally. The smartphone penetration rate exceeds 70%, creating an ideal foundation for the widespread adoption of digital services such as e-commerce, digital finance, online education, and e-government - key enablers for the growth of the consumer-facing digital economy. Crucially, this large, digitally active population supports fast consumer testing and scale for new platform services; if complemented by targeted upskilling and lifelong learning programmes, this human capital pool can be progressively converted into higher-value digital talent for platform engineering and data-science roles.

Third, the rise of leading domestic technology corporations - such as Viettel, VNPT, FPT, CMC, VNG, and MoMo - plays a central role in building the country's digital infrastructure and local platforms. These firms have made strategic investments in critical areas such as data centers, 5G networks, and cloud computing, while simultaneously developing diverse digital ecosystems. Their efforts not only meet domestic market demands but also contribute significantly to the

localization of digital technologies, reducing dependence on foreign platforms and creating opportunities for small and medium-sized enterprises (SMEs) to thrive in the digital environment. These national champions can act as anchors for a domestic platform ecosystem by providing infrastructure, talent pipelines, and B2B demand for local software and services—provided competition and procurement policies are calibrated to encourage domestic value capture.

Fourth, Vietnam's innovation-driven startup ecosystem is expanding rapidly, with over 4,000 startups active in promising sectors such as fintech, edtech, agritech, and e-commerce. The emergence of venture capital funds, national innovation centers, co-working spaces, and government-private sector startup support programs has generated strong momentum for digital innovation in the current era. This nascent ecosystem creates fertile ground for experimentation and niche platform development; with targeted bridging finance, accelerator programmes, and links to corporates and universities, a subset of startups could scale into national platform providers.

Finally, Vietnam is actively strengthening international cooperation in key areas such as technology transfer, institutional development, and digital skills training. The country is participating in regional initiatives like the ASEAN Digital Economy Framework Agreement (DEFA) and is advancing bilateral cooperation programs on data governance, cybersecurity, artificial intelligence (AI), and blockchain technology with strategic partners including South Korea, Japan, and the European Union. Such cooperation can accelerate capability building and provide access to best-practice regulatory models and technical standards; when strategically aligned, these partnerships can shorten the learning curve for domestic firms and amplify the impact of national policy instruments.

5.2. Challenges

Despite encouraging initial achievements, the development of Vietnam's digital economy continues to face a number of systemic barriers that require timely and targeted policy interventions.

First, a major challenge lies in the pronounced disparities in digital development across regions and population groups. While major economic centers such as Hanoi, Ho Chi Minh City, and Da Nang have built relatively mature digital ecosystems, rural areas, mountainous regions, and communities of ethnic minorities still face significant obstacles in accessing broadband infrastructure, digital devices, and basic digital services. These digital divides not only diminish the effectiveness of infrastructure investment but also exacerbate digital inequality, thereby limiting the nationwide spillover effects of the digital economy. The spatial fragmentation also reduces the size of addressable, digitally sophisticated markets for platform firms outside the main agglomerations, thereby constraining the geographic scale and resilience of domestic platforms unless region-targeted interventions and connectivity subsidies are introduced.

Second, the digital skill level of the workforce remains low and uneven, particularly among small and medium-sized enterprises (SMEs), which account for over 90% of all businesses in Vietnam. Many SMEs either lack awareness of the strategic value of digital transformation or face constraints in financial and technical resources to implement it. According to a 2022 survey by the Ministry of Planning and Investment, only about 30% of SMEs had a clear digital transformation strategy, while more than 55% had no plans in place. The shortage of high-quality digital talent, especially in core areas such as big data, cybersecurity, artificial intelligence (AI), and digital product design, has emerged as a critical bottleneck, hindering the development of a robust domestic digital value chain. Without large-scale reskilling, apprenticeship and industry-academic

collaboration programmes, the economy risks perpetuating an “adopter” rather than “innovator” role—where firms implement foreign solutions but fail to build proprietary capabilities.

Third, although Vietnam has made significant progress in establishing a legal framework for the digital economy, the institutional system remains fragmented and slow to adapt to the rapid evolution of new business models. The enactment of key regulations such as the amended Law on Electronic Transactions, the National Data Strategy, and the forthcoming Personal Data Protection Law are important milestones. However, substantial regulatory gaps persist in areas such as platform economy governance, cross-border data management, digital identity systems, the sharing economy, and regulatory sandbox mechanisms. Fragmentation in the state management of digital technology sectors also reduces coordination efficiency and increases compliance burdens for enterprises. Completing VNeID integration and publishing interoperable data standards will reduce onboarding friction, enable secure e-KYC and unlock cross-sector integration for fintech and GovTech applications. This institutional fragmentation raises regulatory uncertainty for investors and entrepreneurs and weakens the capacity to deploy cohesive industrial policy measures that support local platform scaling.

Fourth, investment in research and development (R&D) for digital technology in Vietnam remains modest compared to countries with similar or more advanced levels of development. With total R&D expenditure accounting for only 0.53% of GDP in 2021, Vietnam significantly trails countries like China (2.44%), South Korea (4.81%), and Thailand (1.14%). This structural weakness undermines the country’s ability to master critical technologies and limits the global competitiveness of domestic tech firms. Consequently, without a step-change in R&D financing models—blending public funding, corporate R&D tax incentives, and venture capital to underwrite scale-up—Vietnam will struggle to move from integrator to originator in key technology stacks.

Finally, Vietnam’s heavy dependence on foreign digital platforms across key domains, such as search engines, social media, payment services, and cloud computing, raises serious concerns about data security, digital sovereignty, and the operational cost of national digital systems. In the context of continued market dominance by global tech giants such as Google, Facebook, and Amazon, domestic technology companies face considerable challenges in scaling their operations, attracting users, and competing in core technological capabilities. This dependence increases exposure to external price shocks, reduces local capture of digital value added, and constrains policy space for indigenous ecosystem formation; targeted measures (e.g., localized data infrastructure, procurement leverage, and incentives for domestic cloud and middleware providers) are therefore required to rebalance the ecosystem.

6. Conclusion and policy recommendations

In the context of a world entering a phase of strong transition toward a knowledge-, data-, and digital technology-based economy, developing the digital economy has become a strategic necessity for developing countries, including Vietnam. The digital economy is not merely a new sector, but a method of reorganizing economic and social activities on a digital foundation, promoting productivity, innovation, and enhancing national competitiveness.

In recent years, Vietnam has achieved positive outcomes: the digital economy’s contribution to GDP has grown rapidly, connectivity infrastructure has been expanded, major digital enterprises have emerged, and an initial policy framework for digital transformation has been formed. However, this development process still faces many systemic challenges: digital divides between

regions, a shortage of high-quality human resources, incomplete institutions, and dependence on international technology platforms.

Based on an analysis of the current situation, opportunities, and challenges, this paper proposes several key policy recommendations to promote digital economy development in Vietnam in the coming period, including:

First, improve institutions and policies to support digital economy development. The recently enacted laws are only high-level legal frameworks. To ensure practical effectiveness, Vietnam needs to draft quickly and issue decrees and circulars providing detailed guidance on the implementation of each law. At the same time, technical regulations and industry standards related to data sharing, information security, digital enterprise identification, and clarification of ownership, usage, and transfer of data - especially mixed data (personal, business, and public interest) - should be developed. In addition, a national-level coordination agency on digital transformation and the digital economy should be established with sufficient authority to connect policies across ministries, ensuring data interoperability and effective policy coordination. Establishing a single national coordination body will reduce policy fragmentation, accelerate the issuance of implementing regulations, and ensure coherent sequencing of cross-sector initiatives. In practice, this enables faster deployment of the other instruments recommended in this paper and provides a clear accountability node for monitoring results.

Second, develop modern, synchronized, and secure digital infrastructure. Digital economy development requires upgrading digital infrastructure as an essential infrastructure component. Vietnam needs to: (i) Expand 5G coverage with wide reach, affordable cost, and high speed, especially in industrial parks, innovation centers, and rural areas; (ii) Build a national data center and a network of regional data centers, along with promoting the domestic cloud computing market; (iii) Increase investment in cybersecurity infrastructure and enhance capacity to prevent and respond to cyberattacks; (iv) Support digital infrastructure transformation in localities, especially in disadvantaged provinces, through public investment incentives and support for private participation in providing basic digital infrastructure services. Directing finance toward platform-enabling R&D will address the upstream capability gap identified in the analysis, supporting commercialization of technologies that domestic firms can scale. Over time, this should shift the economy from technology integration toward indigenous platform creation.

Third, develop digital human resources and universal digital skills. Human resources are the key factor ensuring the sustainability of the digital economy. Vietnam needs to build a digital workforce development strategy at three levels: (i) Basic digital skills, by integrating digital literacy content into general education and vocational training programs; (ii) Advanced skills, by expanding the scale and quality of university, postgraduate, and reskilling programs in key areas such as artificial intelligence, big data, blockchain, cybersecurity, and software engineering; (iii) Digital transformation intermediaries, by training a team of personnel capable of evaluating, selecting, and implementing digital technologies, especially for small businesses and local government levels. A public-private digital human resource training support fund should be established, mobilizing participation from major technology enterprises and international organizations in national digital skills development initiatives. A coordinated skills strategy—covering basic literacy, specialist training and intermediaries—will expand the talent pipeline required for platform engineering while widening user capacity to adopt advanced services. This alignment of supply and demand for skills is essential to sustain platform development domestically.

Fourth, promote innovation and develop the domestic digital technology ecosystem. Vietnam should aim to build a digital innovation ecosystem capable of technological autonomy, developing high value-added products, and expanding to international markets. National digital technology zones should be established with special incentives in terms of tax, talent, infrastructure, and data to attract technology enterprises, research institutes, universities, and investment funds. At the same time, promote research and development (R&D) by increasing public spending on digital technology research to at least 1% of GDP by 2030, and encourage private enterprises to invest in laboratories, AI centers, and software testing facilities. Domestic markets should be connected to international markets through strategic cooperation on data, AI, and cybersecurity with partners such as the EU, South Korea, and Singapore, and promote digital service exports. Investing in national and regional data infrastructure and using calibrated procurement levers will lower operating costs for local firms and expand domestic hosting capacity, reducing dependence on foreign providers. This creates the demand and supply conditions necessary for local cloud and middleware providers to scale.

Fifth, narrow the digital divide between regions and ensure equal access. Digital economy development must ensure inclusiveness and access for all citizens. Specific measures include: (i) Implementing commune- and district-level digital transformation support programs focusing on network infrastructure, staff training, and digitalization of essential public administrative services; (ii) Providing public investment incentives and preferential credit for enterprises providing digital services in remote and disadvantaged areas; (iii) Organizing community digital literacy programs, especially for women, the elderly, ethnic minorities, and informal workers; (iv) Developing shared local digital platforms (such as agriculture, market, education, and healthcare data) to support data-driven decision-making by local authorities and businesses. Competitive grants, concessional finance and procurement incentives targeted at lagging provinces will stimulate local supply and demand for digital services and promote geographically inclusive platform markets. Such targeted support helps convert provincial pockets of digital activity into broader national spillovers.

The digital economy presents a major opportunity but also poses high demands on policy coordination, effective public investment, and the proactive adaptability of all stakeholders in society. In the 2025–2030 period, a long-term vision, decisive action, and broad societal consensus are essential for the digital economy to truly become the driving force for Vietnam's rapid and sustainable growth. Sectoral sandboxes and structured procurement pilots provide safe, time-bound environments for firms to test innovations and for regulators to iterate policy, while creating early commercial demand for domestic solutions. Lessons from these pilots should feed directly into scalable regulation and standards.

This study relies primarily on secondary sources and document analysis; therefore, findings are largely inferential rather than causal. Micro-level indicators are not uniformly available, constraining finer-grained analysis. Future work should incorporate representative firm-level surveys, in-depth interviews with platform managers, and policy experiments to evaluate the causal impact of targeted interventions on domestic platform emergence.

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